Computational Physics: Introductory Course

Assignment 3:

Interpolation and curve fitting methods

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What the report should contain

The report should contain:

- The py-files, with carefully commented Python code, that were used to solve the exercises. All code should start with a sentence or two describing what the code is doing. Each py-file should be embedded in the LaTeX or Word document.
- When applicable, you should include the output or plots produced by your code as part of the exercise.
- When you, as part of an exercise, are asked to draw physical conclusions it is important that these conclusions are included in the report. If you are unable to draw any conclusion you should explicitly write 'I was not able to draw any conclusion'.

Important things to keep in mind for producing a good report that adheres to the general rules in scientific writing.

- All variables in the report should be in math font, e.g. the distance was $d = 1.5$ (do not write the distance was d = 1.5, wrong).
- All units in the report should be in correct font, e.g. the acceleration is $a = 1.5$ ms^{-2} (do not write the acceleration is $a = 1.5\,ms^{-2}$, wrong).
- Write $a = 1.5 \cdot 10^{-3}$ instead of $a = 1.5 * 10^{-3}$ (wrong, never use * in equations or text)
- If you give computed values, round them to a reasonable number of decimals.
- The font size in plots should be large enough for the axis texts and legends to be read.

Reading instructions for assignment 3

Start to read chapters 16 and 19 in the lecture notes about interpolation and modeling of data.

Exercises

1. We have the following conversion table for hardness scales from CoroKey, Application guide from Sandvik Coromat, 4:th edition 1997.

OMRÄKNINGSTABELL FÖR HÅRDHETSSKALOR

Inom industrin används många olika system för mätning av materialhårdhet.
I tabellen nedan jämförs de tre vanligaste systemen.

CoroKey skärdatarekommendationer anges i Brinell (HB).

HB 180 för stål (CMC-kod 02.1)

HB 180 för rostfritt stål (CMC-kod 05.21)

HB 260 för gjutjärn (CMC-kod 08.2)

CMC = Coromant materialklassificering (Coromant Material Classification).
Se korsreferenslista för material på sid 4.

Dragbrott-gräns	Vickers	Brinell	Rockwell		Dragbrott-gräns	Vickers	Brinell	Rockwell
N/mm²	**HV**	**HB**	**HRC**	**HRB**	**N/mm²**	**HV**	**HB**	**HRC**
255	80	76,0	–	–	1030	320	304	32,2
270	85	80,7	–	41,0	1060	330	314	33,3
285	90	85,5	–	48,0	1095	340	323	34,4
305	95	90,2	–	52,0	1125	350	333	35,5
320	100	95,0	–	56,2	1155	360	342	36,6
350	110	105	–	62,3	1190	370	352	37,7
385	120	114	–	66,7	1220	380	361	38,8
415	130	124	–	71,2	1255	390	371	39,8
450	140	133	–	75,0	1290	400	380	40,8
480	150	143	–	78,7	1320	410	390	41,8
510	160	152	–	81,7	1350	420	399	42,7
545	170	162	–	85,0	1385	430	409	43,6

Use spline interpolation, see example 16.7 on page 406 in the lecture notes , to determine what Brinell HB-value you get for a tensile failure limit(dragbrottsgräns) of 290 N/mm^2.

2. Fabricate a pendulum using a string and a bob (a weight attached to the string), see figure 1.

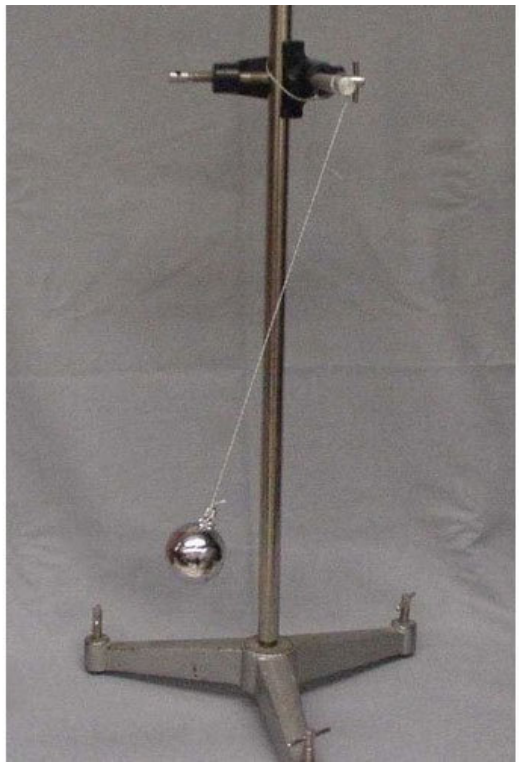


Figure 1: *Pendulum made of a string and a bob attached to a fixed point.*

Measure the period time T for eight different values L of the string. For better accuracy measure the time for 10 oscillations and divide with 10. Don't hold the string between your fingers, but attach it to a fixed point. The oscillation angle should be small.

(a) Plot experimental data as rings.

(b) Figure out if we have a power relation of the form $T = a_0 L^{a_1}$ by plotting with logarithmic scales on both axes.

(c) Fit a non-linear model function $T = a_0 L^{a_1}$ to data.

(d) Plot the fitted model function together with the experimental data.

(e) Use the fitted function to determine how long the string should be to give a period of $T = 1$ s.

3. Position yourself close to a white wall and arrange with some sort of measuring stick or marks (use masking tape) that define a coordinate system in meter (m). A friend throws a ball upwards (vertically) a short distance. You record the throw (use your mobile camera), see figure 2.

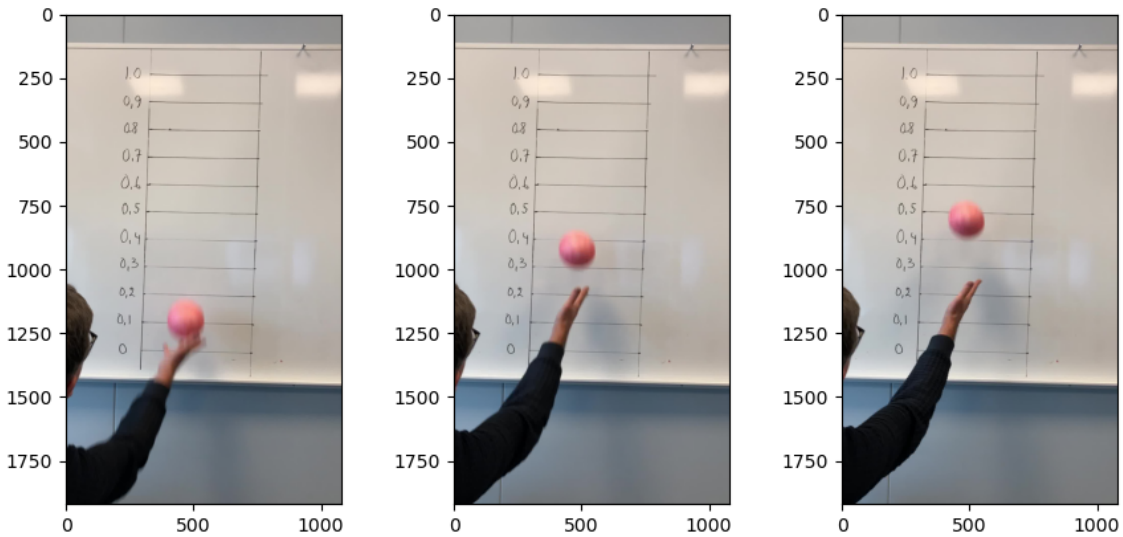


Figure 2: *Throw a ball vertically and record a video. Use some algorithm to determine the row index for the ball. If we have determined the row indexes for $y = 0$ and $y = 1$ in the coordinate system displayed on the whiteboard in the background, we can convert the row index of the ball to the y-coordinate relative to the coordinate system.*

The recorded video will show how the boll moves upwards to its highest point and then moves downwards in an accelerated movement.

Split the video into a sequence of images using some suitable software (see for example `https://ezgif.com/video-to-jpg` or `https://www.videoproc.com/video-editor/convert-video-to-image-sequence.htm`).

Write a program that:

- loops over the images in your video
- opens and stores the image in a matrix B
- uses some algorithm to extract the row index for the position of the ball
- converts the row index of the ball to the y-position relative to the coordinate system and saves the position in a vector *ydata*

After the loop you should create a vector *tdata* with the time in s for each of the images (you need to know the frame rate of your video). Plot *ydata* as a function of *tdata*, it should be a parabola, an perform a least-squares fit to a polynomial of degree 2 see and determine a value of the acceleration g due to gravity. Read section 19.8 in the lecture notes and take inspiration from this. Use the command `polyfit` to perform the least-squares fit to the polynomial of degree 2, see example 19.1 on page 490. Hint: the relation between the row index in the image and the

y-position in the coordinate system is given by

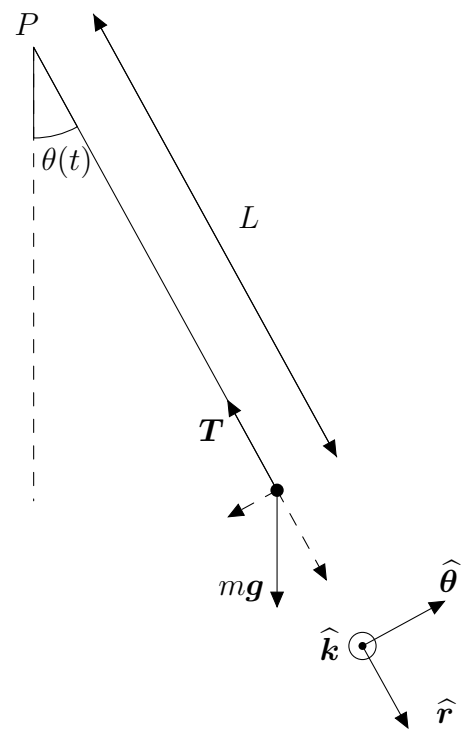
$$y = k \times \text{row index} + m.$$

If we determine the row indexes for, respectively, $y = 0$ and $y = 1$ in our coordinate system, we get a system of two equations, from which we can determine k and m. To design some algorithm to extract the index for the position of the ball we recommend you, on one of the images, to 'hoover' over the ball with the mouse pointer and see what RGB values you get. See how these RGB values differ from the surrounding points. You can now design a criteria where you detect the typical RGB values of the ball. If you are unable to make an automated detection system you can manually loop over all the images in your video and place the mouse pointer at the ball and write down, on a sheet of paper, the row index you got. Then, in your program, use these values, converted to y-positions relative to the coordinate system as elements in the *ydata* vector. On Canvas we will put a short video showing how you can 'hoover' over points in an image with the mouse pointer to see both the RGB values as well as the row and column indexes.

Physics background

The course is in computational physics, and we provide the physics background for many of the exercises together with links, that make it possible to read more if you find it interesting.

Physical pendulum
The coordinate system and force diagram for the simple pendulum is shown in the figure below.



The vectors $\widehat{\boldsymbol{r}}$, $\widehat{\boldsymbol{\theta}}$, $\widehat{\boldsymbol{k}}$, with $\widehat{\boldsymbol{k}}$ pointing out of the plane of the figure, define a right handed orthonormal co-ordinate system. The torque (sv. vridmoment) about the pivot point P is given by

$$\boldsymbol{\tau} = L\widehat{\boldsymbol{r}} \times m\boldsymbol{g},$$

where $m\boldsymbol{g}$ is the force due to gravity. Looking at the figure we see that the latter can be written

$$m\boldsymbol{g} = mg\left(\cos(\theta)\widehat{\boldsymbol{r}} - \sin(\theta)\widehat{\boldsymbol{\theta}}\right).$$

The rules for vector product give

$$\boldsymbol{\tau} = -Lmg\sin(\theta)\widehat{\boldsymbol{k}}.$$

The moment of inertia of a point mass about the pivot point P is

$$I = mL^2.$$

The rotational equation of motion is then

$$\tau_z = I\alpha,$$

where τ_z is the component of the torque along $\widehat{\boldsymbol{k}}$ and $\alpha = \theta''(t)$ is the angular acceleration. Inserting the expression for the z component of the torque and rearranging gives

$$\theta''(t) = -\frac{g}{L}\sin(\theta(t)).$$

In the small angle approximation $\sin(\theta) \approx \theta$ this reduces to

$$\theta''(t) = -\frac{g}{L}\theta(t).$$

The solution of the latter equation is given by

$$\theta(t) = A\cos(\omega t) + B\sin(\omega t),$$

with $\omega = \sqrt{g/L}$, from which we get the period time

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{L}{g}}.$$

Vertical motion of a ball

We will consider the vertical motion of a ball. Assuming a co-ordinate system oriented upwards the acceleration is

$$y''(t) = -g,$$

where $g \approx 9.81\ \mathrm{ms}^{-2}$ is the local gravitational acceleration. Integrating we have

$$y'(t) = v_0 - gt,$$

where v_0 is the vertical velocity of the ball at $t = 0$. Integrating once more we get

$$y(t) = y_0 + v_0 t - \frac{g}{2}t^2,$$

where y_0 is the vertical position of the ball at $t = 0$. Measuring the position $y(t)$ at a number of times and performing a least-squares fit to a polynomial of degree two allows us to extract y_0, v_0 and g.